Biostatistics 'B' Advanced course

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the problem is the noise

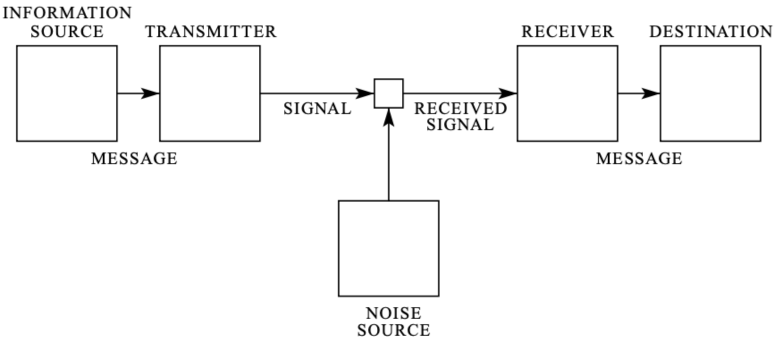
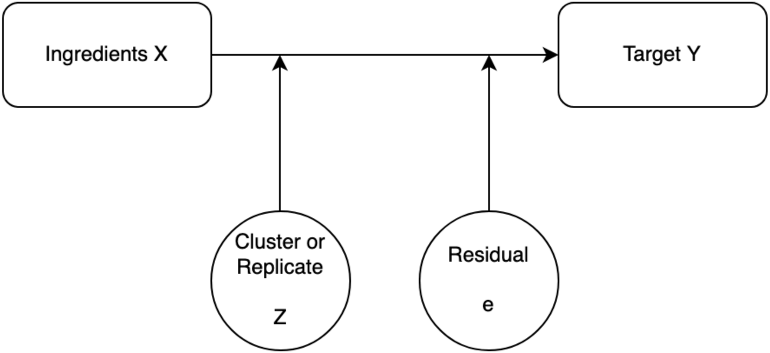


Fig. 1 — Schematic diagram of a general communication system.

the art of noise /2



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Commentary

The Table 2 Fallacy: Presenting and Interpreting Confounder and Modifier Coefficients

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stop making noise!

Table 2. Cause-specific Hazard Ratios for Use of Psychotropics in Association

	Antidepressants	
	HR[a]	95% CI
Work-related violence (yes vs. no)	1.38	1.09, 1.75
Women vs. men	1.41	1.18, 1.68
Age per 5-year increase	1.01	0.98, 1.06
Cohabitation (yes vs. no)	0.81	0.67, 0.97
Education per SD increase	0.88	0.81, 0.96
Income per quartile increase	0.91	0.83, 0.99
Social support from colleagues per unit increase	0.95	0.87, 1.05
Social support from supervisor per unit increase	0.95	0.87, 1.03
Influence per unit increase	0.93	0.86, 1.02
Quantitative demands per SD increase	1.02	0.94, 1.11

stop making noise!

ASHA American Speech-Language-Hearing Association

JSLHR

Tutorial

Mediators, Moderators, and Covariates: Matching Analysis Approach for Improved Precision in Cognitive-Communication Rehabilitation Research

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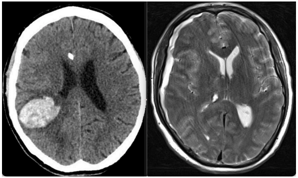
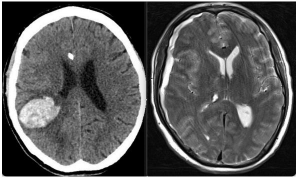
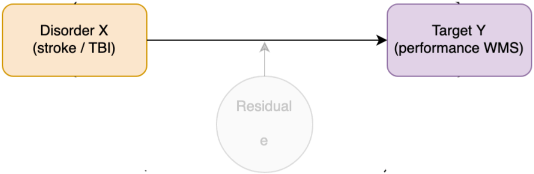


- to assess whether a given treatment is effective in improving memory performance after stroke or traumatic brain injury
- to focus on target variable, one might treat other individual characteristics (e.g. demographic - sex, age - or behavioral - exercise, sleep -) as noise by removing their associated variability from the statistical model
- This gives researchers an idea of whether a treatment works at the *group level* for an "average person".



WMS-IV
Flexible Approach
PEARSON



Disorder X
(stroke / TBI)
Target Y
(performance WMS)
Residual
e

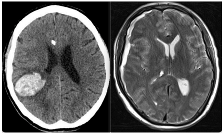
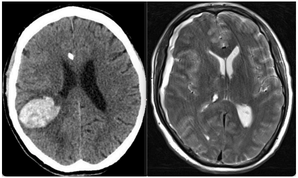


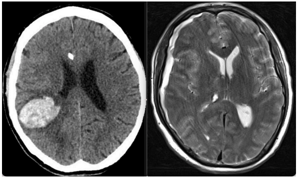
Tabella: Independent Samples T-Test

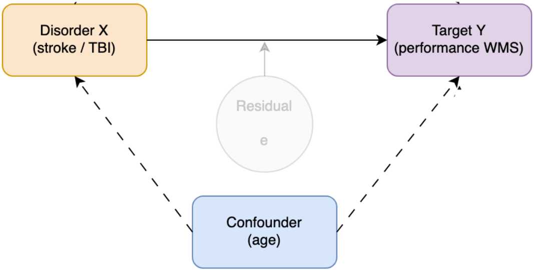
Test	Statistic	df	p
Student	4.097	198.000	$< .001$
Welch	4.097	197.526	$< .001$
Mann-Whitney	6523.500		$< .001$

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	15.293
H_1	0.280	0.078	0.074	14.720



- a **confounder** is causally associated with both the independent and dependent variables but does not drive the association between them
- increasing age is associated with both an increased risk of traumatic brain injury and stroke
- however, traumatic brain injury or stroke does not affect a person's age

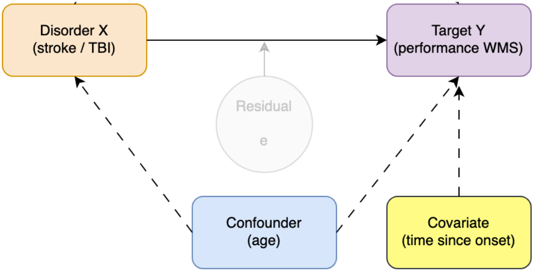



Disorder X
(stroke / TBI)
Target Y
(performance WMS)
Residual
e
Confounder
(age)

- a **covariate** affects the outcome variable but is not related to the independent variable
- time since onset may affect a persons performance on a memory outcome (Y), but it is not causally associated with the presence or absence of a traumatic brain injury or stroke (X)


MARC FEBRUARY

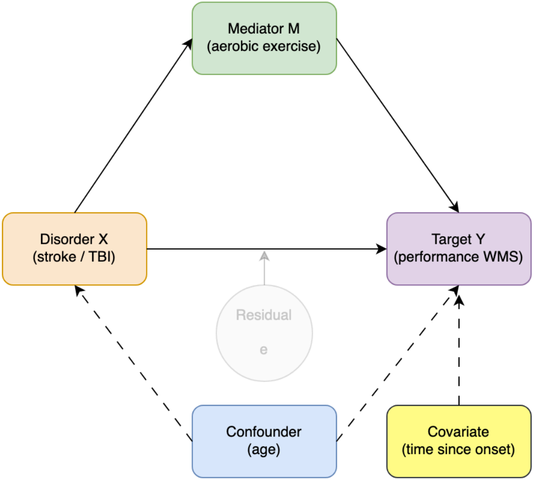

WMS-IV


Disorder X
(stroke / TBI)
Target Y
(performance WMS)
Residual
e
Confounder
(age)
Covariate
(time since onset)

- the amount of exercise a person gets in a week might affect memory performance
- in a standard linear model, one might adjust for exercise as a covariate to assess whether stroke/tbi affects the target (memory performance) *without* the effects of exercise
- however, what if exercise is, in fact, a key part of the picture?
- maybe stroke/tbi causes people to exercise less, and this reduction in exercise affects memory performance. In this case, exercise is a **mediator** which lies on the **causal pathway**.

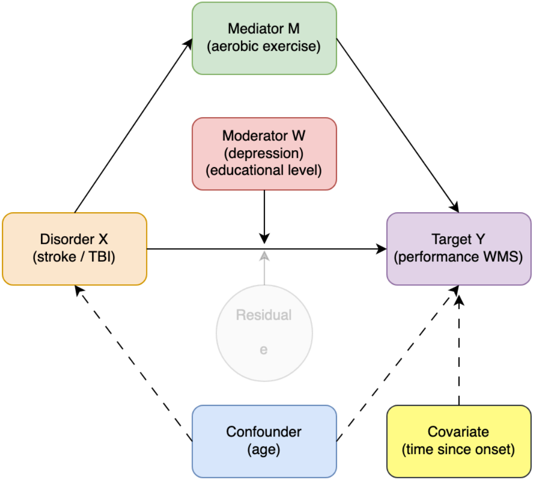

28-DAY HOME WORKOUT CHALLENGE
DAY 16
• Camel pose 50 sec
• Dynamic downward dog 35 sec
• Plank 40 sec
• Goddess squat 60 sec


WMS-IV


Mediator M
(aerobic exercise)
Disorder X
(stroke / TBI)
Target Y
(performance WMS)
Residual
e
Confounder
(age)
Covariate
(time since onset)

- depression can also affect memory
- a **moderator** is not on the causal pathway but interacts with the independent variable in a way that influences the outcome
- a moderation model differs from controlling or adjusting for a covariate in that it asks what effect a treatment has on outcomes *at different levels of the moderator*, whereas adjusting for a covariate asks what effect the treatment would have on outcome *if the value of the covariate was held constant*
- moderators are key to understand context


WMS-IV
WMS-IV


Mediator M
(aerobic exercise)
Moderator W
(depression)
(educational level)
Disorder X
(stroke / TBI)
Target Y
(performance WMS)
Residual
e
Confounder
(age)
Covariate
(time since onset)

mediator or moderator?

- a mediator lies on the causal pathway between X (treatment) and Y (the target or outcome)
  - importantly, a mediator must vary with the independent variable
- a moderator affects the relationship between X and Y (e.g., changes the magnitude or direction of the effect) but does not form part of the causal chain linking them

Stable characteristics such as sex or ethnicity are never mediators, as they cannot be caused by or vary with the independent variable, but they might be important moderators. Constructs such as behaviors, symptoms, and functional status often are potential mediators, but they could be moderators as well.